

MD SAMIUL ALAM SUMEL

AI Product Engineer — Port & Logistics Domain

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PROFESSIONAL SUMMARY

AI Product Engineer with 12+ years of hands-on experience in cargo billing, vessel/vehicle tracking, and port authority workflows at Mongla Port Authority. Finds the operational problem, designs the system, builds it with AI-assisted development, then rebuilds each piece independently from memory before counting it as learned. 3 self-built tools — billing, vehicle tracking, payroll automation — are in daily production use. Skills backed by that shipped software and self-directed study (The Odin Project, Full Stack Open), not a bootcamp certificate. Builds frontend independently; backend/data work with AI-assisted development (Claude Code), closed through a self-authored engineering roadmap. Not a bootcamp graduate or senior engineer, but bringing 12+ years of professional discipline from a demanding operational job. Seeking a remote AI Product Engineer, Port Operations Technologist, Logistics Systems Engineer, or full-stack role, worldwide, with no time zone constraints.

TECHNICAL SKILLS

Domain & Port Operations: Port billing & tariff systems, wharfrent/VAT/levy calculation, cargo & vessel tracking, C&F agent workflows, customs/regulatory compliance context

Frontend & JavaScript: HTML5 (Semantic), CSS3, Responsive Design, Tailwind CSS, DaisyUI, JavaScript (ES6+), DOM & Events, Fetch & Promises / Async-Await, Array/Object Methods, ES Modules

React & Next.js: Own: React (Hooks, Context), React Router, React Hook Form. AI-assisted: TanStack Query, Axios (incl. Interceptors), Next.js App Router, NextAuth

Backend, Data & Deployment: Own: Git & GitHub (Branching, PRs), deployment (Cloudflare Pages/Firebase/Netlify). AI-assisted: Node.js & Express, REST APIs (CRUD), MongoDB (Atlas), Aggregation Pipeline, JWT Auth, Firebase Auth & Admin SDK, Next.js API Routes, Stripe Payments, Pagination & Search

Linux & Automation (self-directed): Bash Scripting, Linux Administration (Ubuntu), systemd, Cron & Timers, Monitoring & Alerting Scripts

PROFESSIONAL EXPERIENCE

Senior Outdoor Assistant — Port Operations & IT Systems

Nov 2017 – Present

Mongla Port Authority — Bangladesh's second-largest international seaport

Core Port Operations

- Manage end-to-end wharfrent billing cycle — slab-wise charge computation, VAT and levy calculation, inside/outside cargo split, and final billing issuance for active C&F agents; identified the recurring manual-calculation bottleneck and independently built and deployed portbill, now used daily by the billing desk.
- Track cargo and vehicle positions across warehouse, shed, and yard terminal locations in real time; the single-person, Excel-only version of this record was the direct trigger for building carview, now the Traffic Department's shared, version-controlled tracking system.
- Operate the port automation billing system end-to-end and coordinate C&F agent documentation and advance payment planning.

Self-Initiated Web Development & Learning

- Independently identified critical inefficiencies in manual paper-based billing, overtime payroll calculation, and vehicle tracking workflows — designed and deployed 3 live web applications now used daily by C&F agents and port staff, built with plain JavaScript/HTML/CSS and AI-assisted development as a productivity tool.
- Worked through a large part of a full-stack curriculum (HTML/CSS, JavaScript, React, part of Node.js/Express and Next.js) alongside a full-time port operations career, before stepping away to focus on shipping real projects.
- Maintain an active public GitHub portfolio (github.com/samiulAsumel) demonstrating continuous technical growth.

Junior Outdoor Assistant — Port & Terminal Operations

Nov 2013 – Nov 2017

Mongla Port Authority

- Assisted in port inspections, vessel operations, and cargo handling under senior supervision; monitored cargo dwell time and vehicle positioning across terminal locations.
- Built foundational expertise in wharfrent billing, maritime logistics, and customs clearance that underpins all later self-initiated digital work.

FEATURED PROJECTS

Port Billing Calculator — [Live](#) | [GitHub](#)

Problem: Port billing system creates permanent entries — C&F agents had no way to estimate wharfrent charges before committing, causing billing disputes, excessive dwell time, and repeated counter visits.

Real-time advance wharfrent calculator: slab-based charge computation, VAT and levy calculation, inside/outside cargo split, hoisting charge auto-calculation, print-ready A4 output. Actively used by C&F agents at Mongla Port daily.

Tech: JavaScript, HTML/CSS, Cloudflare Pages, PWA, Print Output

Daily Car Balance & Location Tracking System ("carview") — [Live](#) | [GitHub](#)

Problem: Vehicle positions across warehouse, shed, and yard were recorded in one person's personal Excel file — invisible to all other staff, creating bottlenecks and dependency on a single individual.

Offline-first PWA tracking 8 port locations, with Cloudflare Worker + private GitHub repo sync, a Chart.js analytics dashboard (7 charts + KPIs), 13 report sections, and Excel export. Built for Mongla Port Authority's Traffic Department.

Tech: PWA, Cloudflare Worker, Chart.js, GitHub Sync, Excel Export

OT Bill Management System — [Live](#) | [GitHub](#)

Problem: Overtime billing was a fully manual Excel process — multi-step hourly rate calculations and final bill generation done by hand every cycle.

Staff enters an employee profile once; the system generates the complete final OT bill instantly with correct hourly rate, cumulative date-wise OT calculation, and A4 print-ready output. Reduced billing cycle time from hours to minutes.

Tech: JavaScript, Employee DB, Auto-Calc, A4 Print, Cloudflare Pages

Client Intake Form — [Live](#) | [GitHub](#)

Problem: Collecting project requirements from a client over email or chat is unstructured — details get missed, wasting time on both sides.

Single-file, client-side project requirement intake form — structured questions, no backend required, mailto-based report so the completed brief lands directly in an inbox.

Tech: JavaScript, HTML/CSS, Client-Side Only

World Kitchen Atlas — [Live](#) | [GitHub](#)

Next.js static-export culinary encyclopedia on Cloudflare Pages, with a Cloudflare Worker proxying a private GitHub data repo, admin CRUD, and a Durable-Object-backed visit counter.

JARVIS (source available, no public demo) — [GitHub](#)

Self-hosted personal AI assistant with multi-model routing (Groq, OpenRouter, Claude, Gemini, local Ollama), voice I/O, and persistent memory.

LINUX / AUTOMATION PRACTICE

Self-directed practice building production-style Bash automation for deployment (auto-deploy-pipeline), SSL certificate renewal (auto-ssl-manager), and log/security monitoring (log-analyzer-alert). Daily Linux (Ubuntu) user; self-driven learning, not job-critical production work.

SELF-DIRECTED STUDY

The Odin Project & Full Stack Open

Aug 2026 – Present

Self-Directed — In Progress

Self-paced study for code comprehension, not a certificate — starting from The Odin Project's Intermediate HTML and CSS course, then Full Stack Open (University of Helsinki): JavaScript fundamentals, Node.js/Express in depth, testing, and Git workflow, built from assignment briefs rather than a course guide.

EDUCATION & LANGUAGES

Higher Secondary Certificate — Business Studies: Khulna Public College, 2009 – 2011

Secondary School Certificate — Business Studies: Mongla Bandar Secondary School, 1999 – 2009

Languages: Bengali — Native | English — Professional